Functions (Part 2)

PreCalculus - I

26CS(Evening) - Fall 2026

Function Arithmetic

For functions $f, g : A \to \mathbb{R}$, the sum, difference, product and quotient are defined on A by

$$\begin{aligned}(f+g)(x) &= f(x) + g(x)\\ (f-g)(x) &= f(x) - g(x)\\ (f\cdot g)(x) &= f(x)\cdot g(x)\\ (f/g)(x) &= f(x)/g(x), \quad g(x) \neq 0\end{aligned}$$

Function Arithmetic: Examples

Example 1: $f_1(x) = 2x,\ f_2(x) = x^2$.

$$(f_1 + f_2)(x) = 2x + x^2, \qquad (f_1 \cdot f_2)(x) = 2x^3.$$

Example 2: $f(x) = 3x + 2,\ g(x) = -2x + 1$.

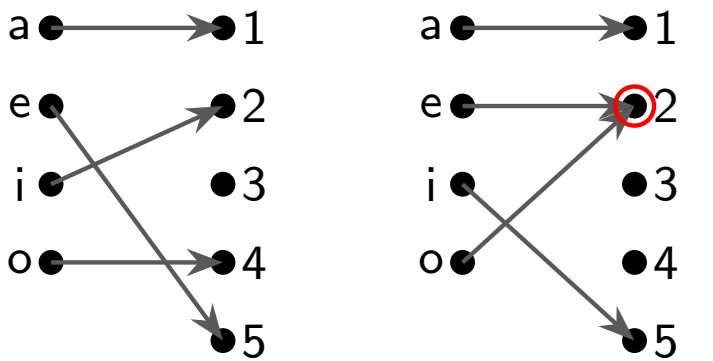
$$(f \cdot g)(x) = (3x + 2)(-2x + 1) = -6x^2 - x + 2.$$

At $x = -1$: $f(-1) = -1,\ g(-1) = 3$, so $(f \cdot g)(-1) = -3$ (matches direct substitution).

One-to-One (Injective) Functions

Definition: Each codomain element has a unique pre-image:

$$\boxed{f(a) = f(b) \Rightarrow a = b} \quad \text{equivalently} \quad a \neq b \Rightarrow f(a) \neq f(b).$$



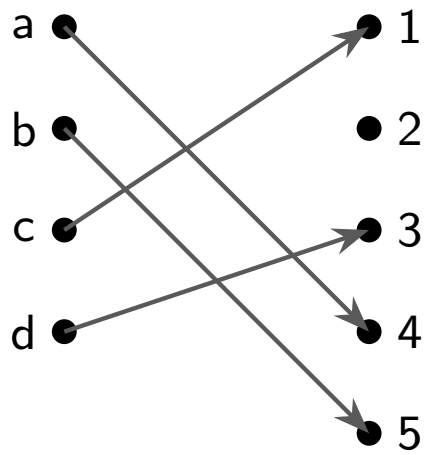
One-to-one

Not one-to-one

"Injective" = "injection" = one-to-one; unused codomain elements are allowed.

One-to-One: Example

Is $f : \{a, b, c, d\} \to \{1, 2, 3, 4, 5\}$ with
$f(a) = 4,\ f(b) = 5,\ f(c) = 1,\ f(d) = 3$ one-to-one?

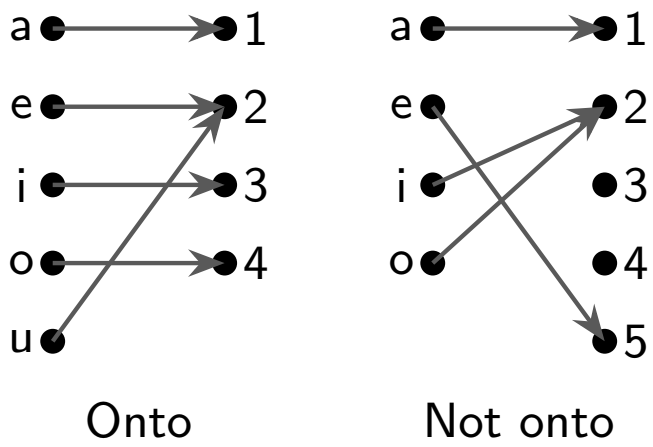


Yes — all images are distinct.

Onto (Surjective) Functions

Definition: Every codomain element is an image of some pre-image:

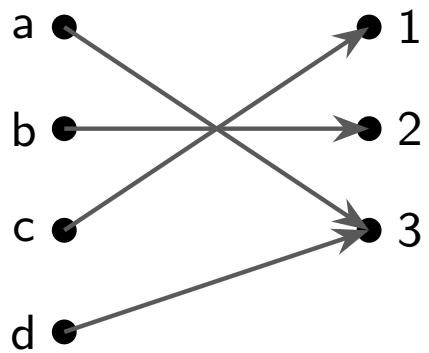
$$\forall b \in B,\ \exists a \in A : f(a) = b.$$



"Surjective" = "surjection" = onto; multiple pre-images per codomain element are allowed.

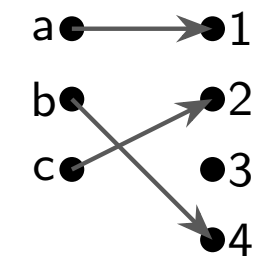
Onto: Example

Is $f : \{a, b, c, d\} \to \{1, 2, 3\}$ with
$f(a) = 3,\ f(b) = 2,\ f(c) = 1,\ f(d) = 3$ onto?

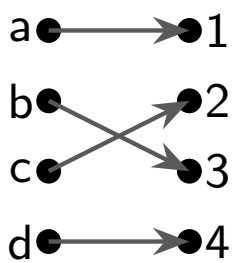


Yes — every codomain element is used.

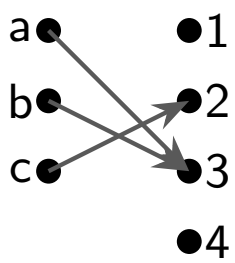
Onto vs. One-to-One: Quick Test



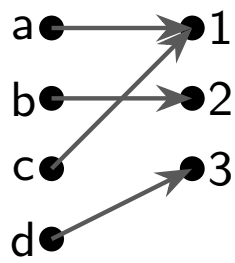
1-to-1, not onto



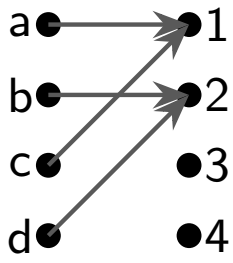
Both



Not a function



Onto, not 1-to-1



Neither

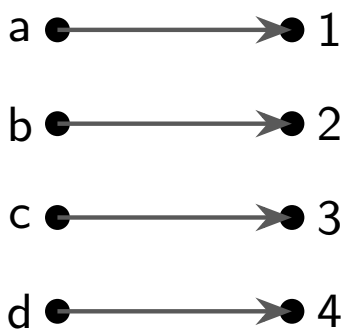
Examples: Proving 1-to-1 / Onto

$f(x) = x^2$ **on** $\mathbb{Z}$ **is not 1-to-1:** if $x_1^2 = x_2^2$, then $(x_1 + x_2)(x_1 - x_2) = 0$, so $x_1 = -x_2$ or $x_1 = x_2$. Since $x_1 = -x_2$ is possible with $x_1 \neq x_2$ (e.g. $2, -2$), f is not 1-to-1.

$f(x) = x + 1$ **on** $\mathbb{Z}$ **is onto:** given $y \in \mathbb{Z}$, take $x = y - 1 \in \mathbb{Z}$. Then $f(y - 1) = (y - 1) + 1 = y$. Hence onto. ✓

Bijections

Bijection = both one-to-one and onto (a one-to-one correspondence).



Examples:

- $f(x) = x^3$ on $\mathbb{R}$: 1-to-1 (strictly increasing) and onto (real cube root) $\Rightarrow$ bijective.
- $f(x) = \dfrac{x+1}{x}$ on $\mathbb{R} \setminus \{0\}$: 1-to-1, but 1 is never attained ($x + 1 = x$ impossible) $\Rightarrow$ not bijective.

Floor, Ceiling, Identity

Floor: $\lfloor x \rfloor$ = largest integer $\leq x$; **Ceiling:** $\lceil x \rceil$ = smallest integer $\geq x$.

x	$\lfloor x \rfloor$	$\lceil x \rceil$
2.4	2	3
2.7	2	3
-2.7	-3	-2
-2	-2	-2

Identity function: $f(x) = x$ for all $x \in A$; domain and codomain must be the *same* set.

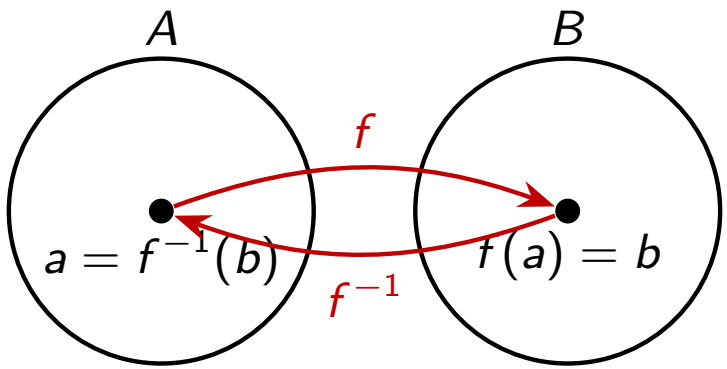
Note

Image and pre-image always coincide under the identity.

Inverse Function

If f is a bijection from A to B, its inverse $f^{-1} : B \to A$ is defined by

$$\boxed{f^{-1}(b) = a \iff f(a) = b.}$$



f invertible $\iff$ f is a bijection. Inverse is *not* defined if f is not 1-to-1 (multiple pre-images) or not onto (some b has none).

Inverse: Examples

Example 1: $f : \{a, b, c\} \to \{1, 2, 3\}$, $f(a) = 2$, $f(b) = 3$, $f(c) = 1$.
This is bijective, so invertible:

$$f^{-1}(1) = c, \quad f^{-1}(2) = a, \quad f^{-1}(3) = b.$$

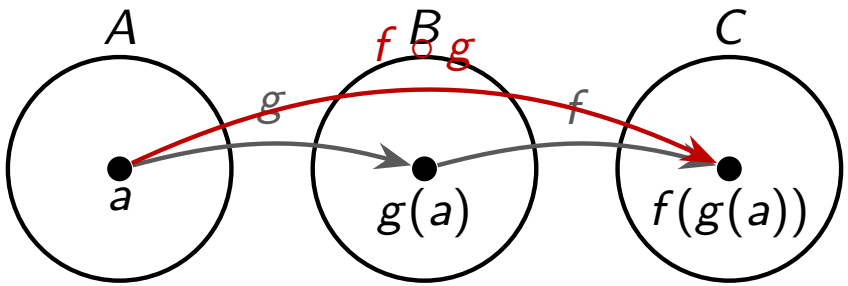
Example 2 (working rule): $f(x) = x + 1$ on $\mathbb{Z}$. Solve $x + 1 = y \Rightarrow x = y - 1$, so

$$f^{-1}(y) = y - 1.$$

Composition

For $g : A \to B$ and $f : B \to C$,

$$\boxed{(f \circ g)(a) = f(g(a)).}$$



Example: $f(x) = 2x + 3$, $g(x) = 3x + 2$.

$$f(g(x)) = 2(3x + 2) + 3 = 6x + 7, \qquad f(g(1)) = 13.$$

Composition: Properties and Worked Table

Not commutative: with $f(x) = 2x + 3,\ g(x) = 3x + 2$,

$$f(g(x)) = 6x + 7, \qquad g(f(x)) = 3(2x + 3) + 2 = 6x + 11,$$

so $f \circ g \neq g \circ f$.

Worked table: $A = \{1, \ldots, 5\}$, $f = \{(1,3), (2,5), (3,3), (4,1), (5,2)\}$, $g = \{(1,4), (2,1), (3,1), (4,2), (5,3)\}$.

$f \circ g$	$g \circ f$
$(f \circ g)(1) = f(4) = 1$	$(g \circ f)(1) = g(3) = 1$
$(f \circ g)(2) = f(1) = 3$	$(g \circ f)(2) = g(5) = 3$
$(f \circ g)(3) = f(1) = 3$	$(g \circ f)(3) = g(3) = 1$
$(f \circ g)(4) = f(2) = 5$	$(g \circ f)(4) = g(1) = 4$
$(f \circ g)(5) = f(3) = 3$	$(g \circ f)(5) = g(2) = 1$

Composition: When It's Not Defined

$g : A \to A$ with $A = \{a, b, c\}$, $g(a) = b$, $g(b) = c$, $g(c) = a$.
$f : A \to B$ with $B = \{1, 2, 3\}$, $f(a) = 3$, $f(b) = 2$, $f(c) = 1$.

Compute $f \circ g$:

$$(f \circ g)(a) = f(b) = 2, \quad (f \circ g)(b) = f(c) = 1, \quad (f \circ g)(c) = f(a) = 3.$$

$g \circ f$ is not defined: range of f is not contained in the domain of g.

Summary

Concept	**Key Idea**
Function $f : A \to B$	Exactly one B-element per $a \in A$
Arithmetic	$(f + g)(x) = f(x) + g(x)$, etc.
One-to-one (injective)	$f(a) = f(b) \Rightarrow a = b$
Onto (surjective)	$\forall b \in B,\ \exists a : f(a) = b$
Bijection	Both one-to-one and onto
Floor / Ceiling	$\lfloor x \rfloor \leq x \leq \lceil x \rceil$
Inverse f^{-1}	Exists only when f is a bijection
Composition $f \circ g$	$(f \circ g)(x) = f(g(x))$, not commutative

Practice Problems

1. $f : \{a, b, c, d\} \to \{1, 2, 3, 4, 5\}$ with $f(a) = 2,\ f(b) = 1,\ f(c) = 4,\ f(d) = 3$: is f 1-to-1? Onto?
2. Is $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x + 1$, 1-to-1? Onto?
3. Show that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$, is a bijection.
4. Compute $\lfloor 3.7 \rfloor,\ \lceil 3.7 \rceil,\ \lfloor -1.4 \rfloor,\ \lceil -1.4 \rceil$.
5. If $f(x) = 4x - 5$, find $f^{-1}(y)$.
6. Let $f(x) = x^2 + 1,\ g(x) = 2x - 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.
7. True/False: A function can be onto but not one-to-one. Give an example if true.

Exercise: Chapter 2, Topic 2.3 — Q 1, 2, 8–12, 22, 23, 36, 37